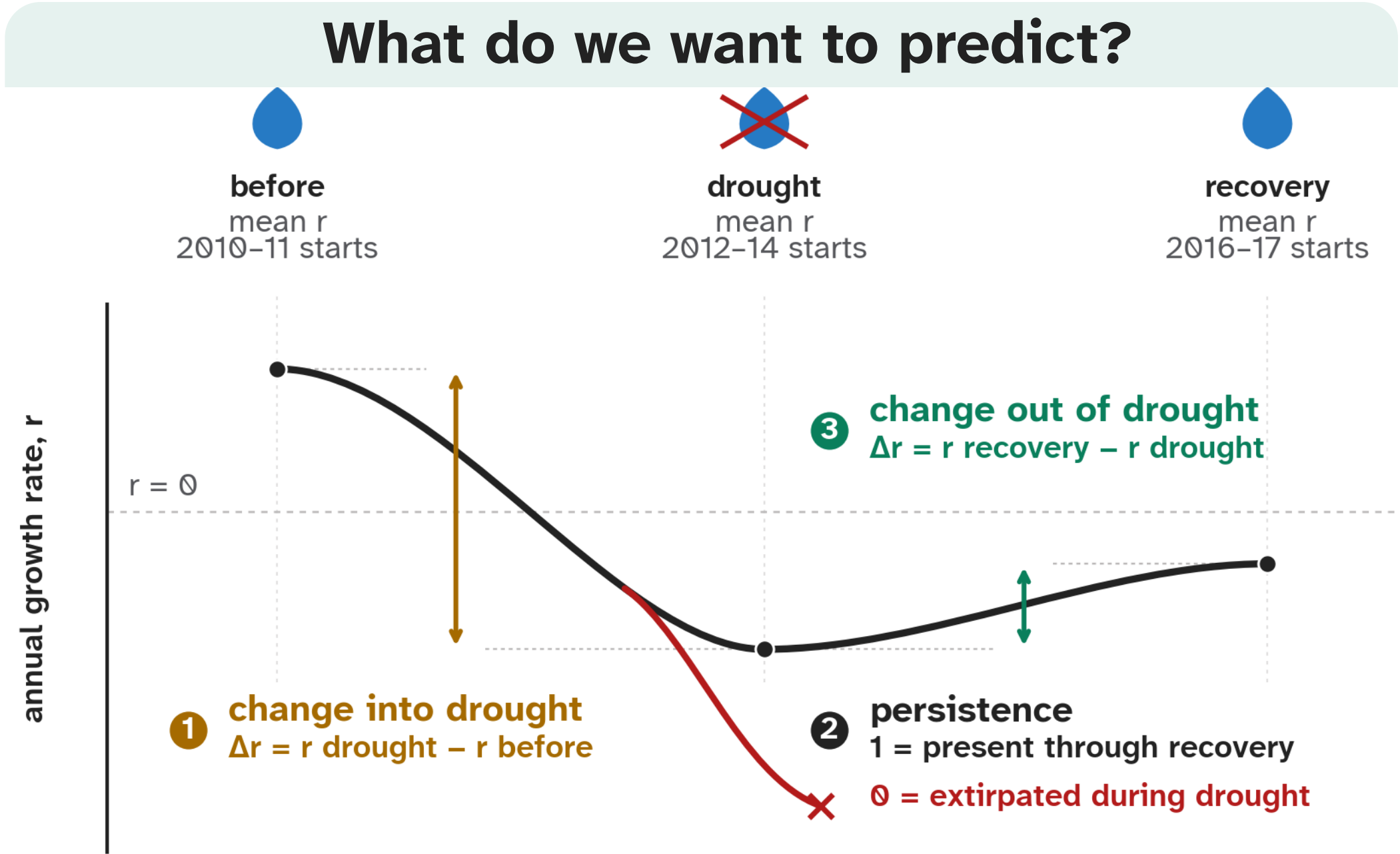
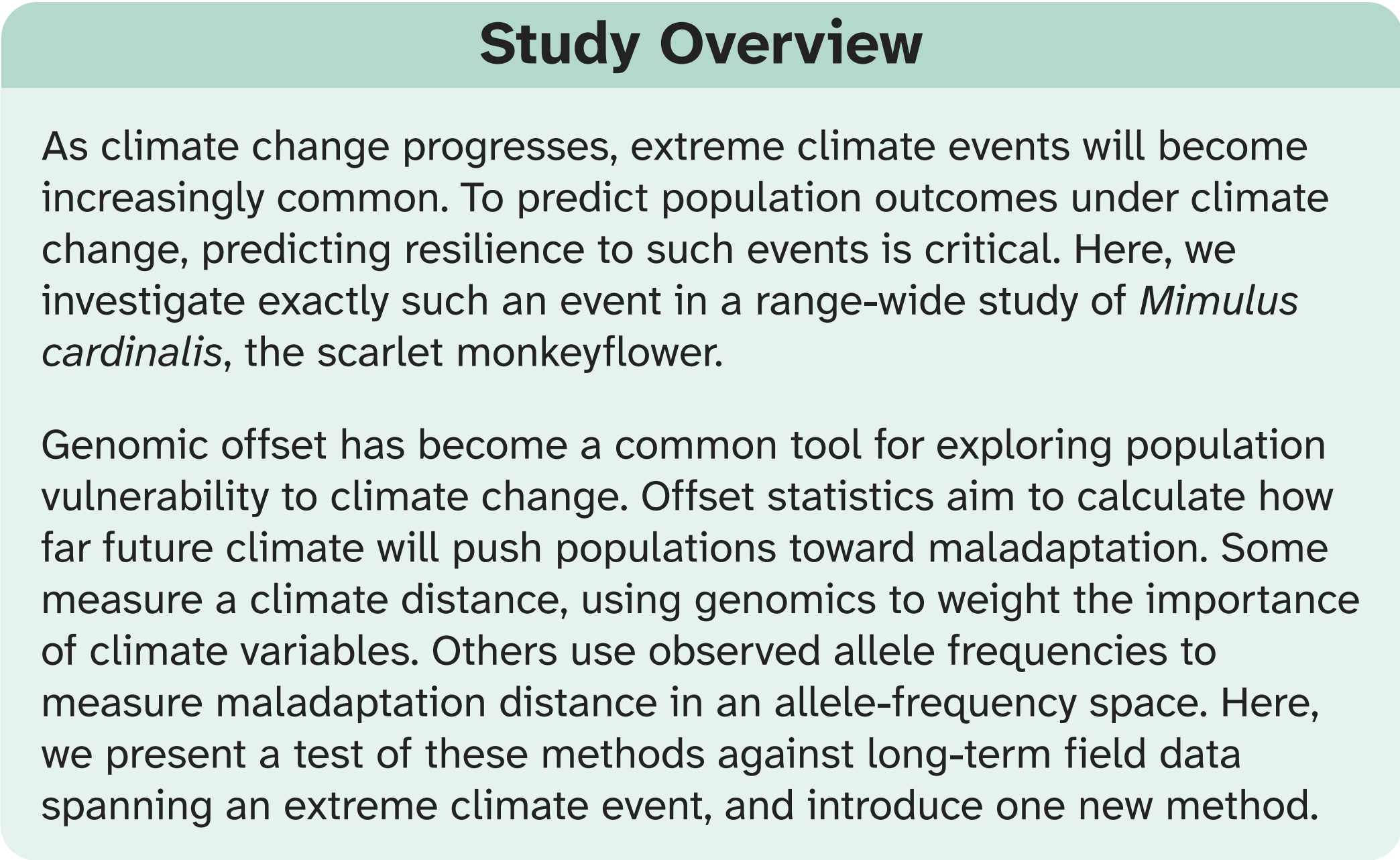
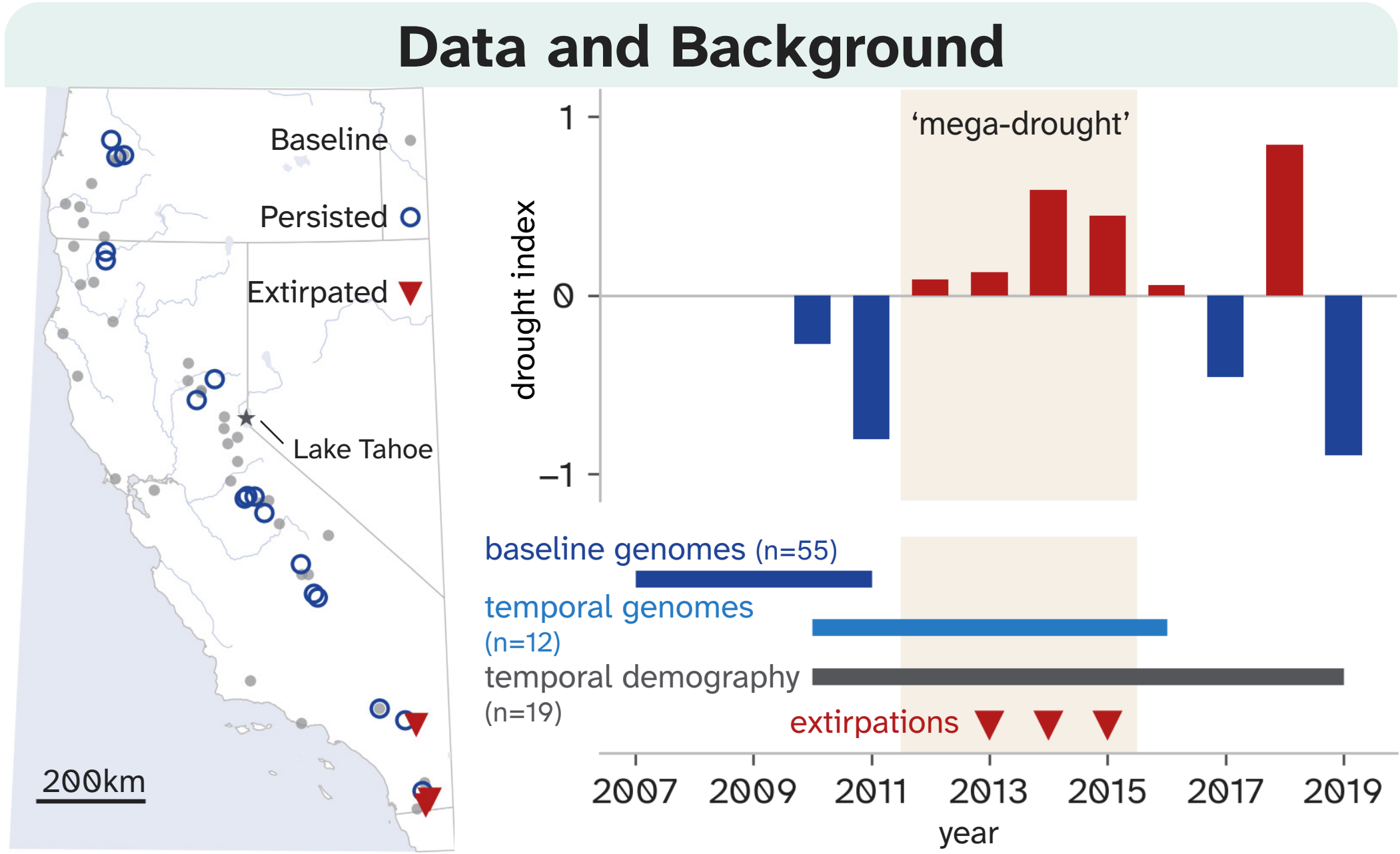


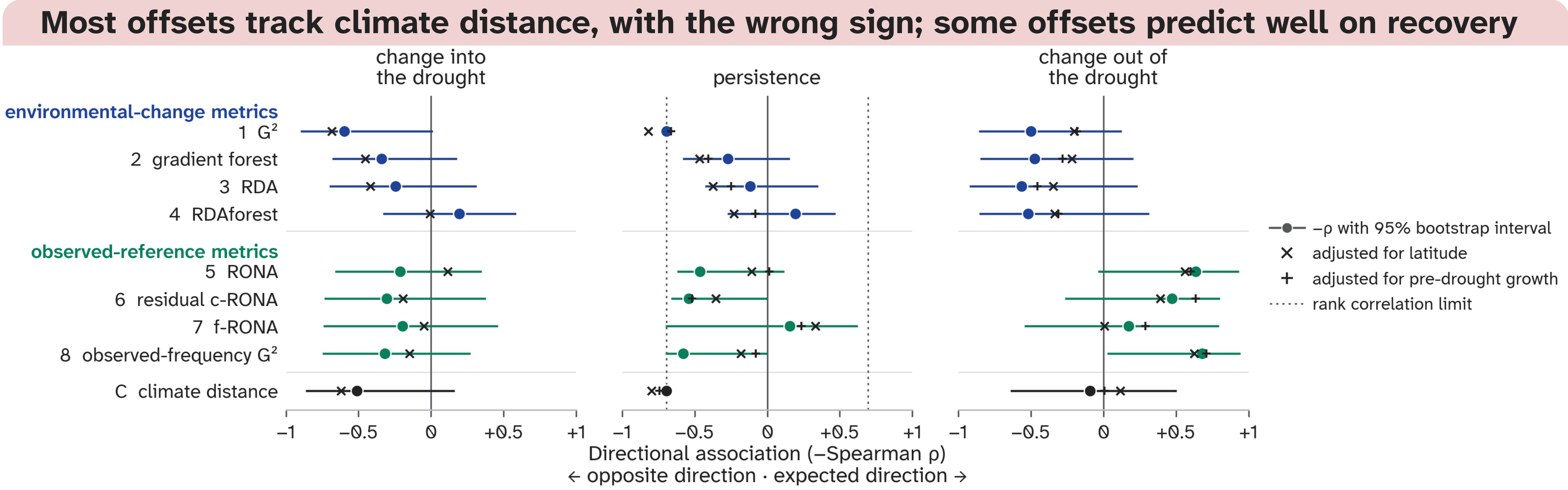
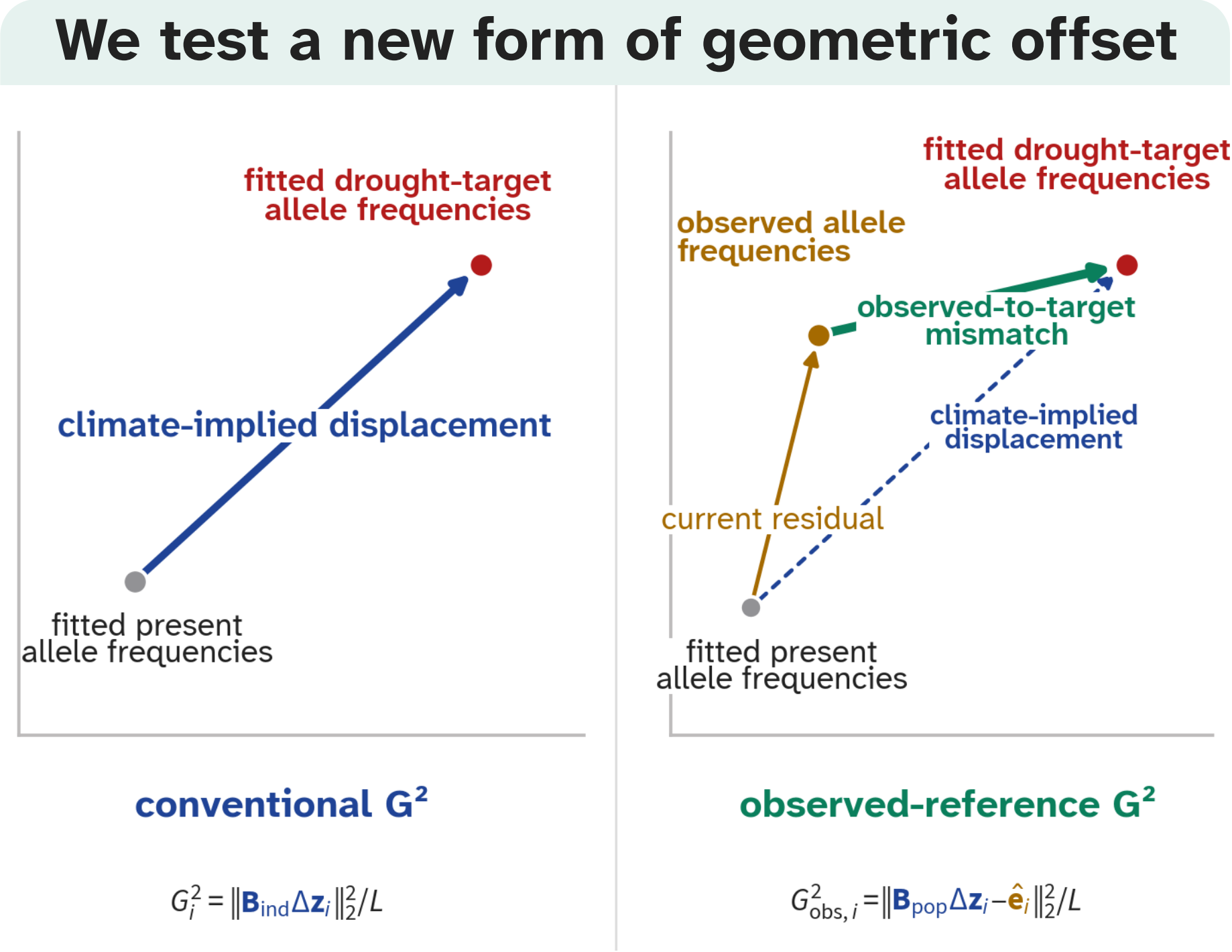
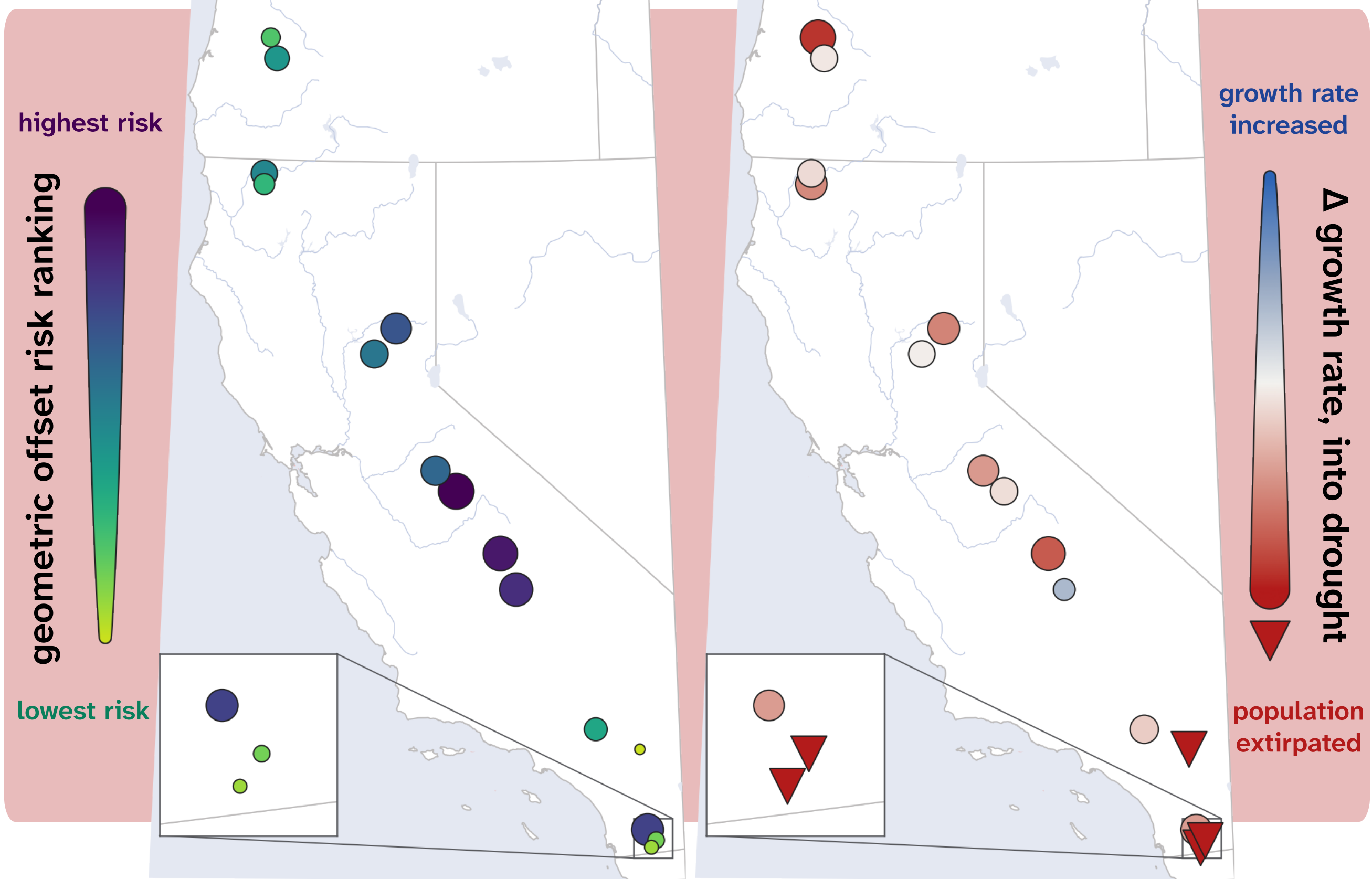
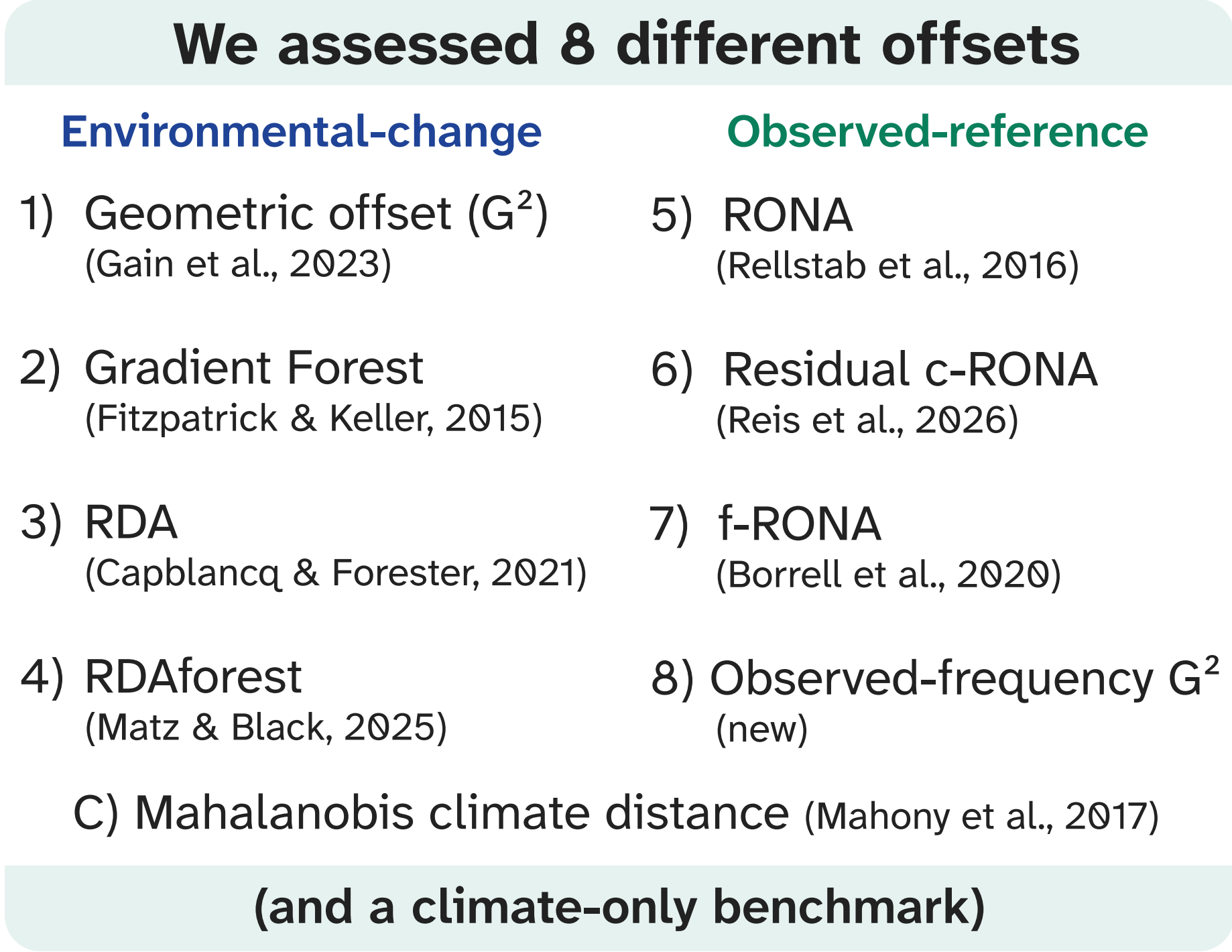
Genomic offset predicts population vulnerability in an extreme climate event... backwards

Does genomic offset predict population resilience to an observed extreme climate event in scarlet monkeyflower?
Christopher A. Talbot¹, Daniel N. Anstett², Philipp W. Messer¹

¹Department of Computational Biology, Cornell University ²School of Integrative Plant Science, Plant Biology Section, Cornell University

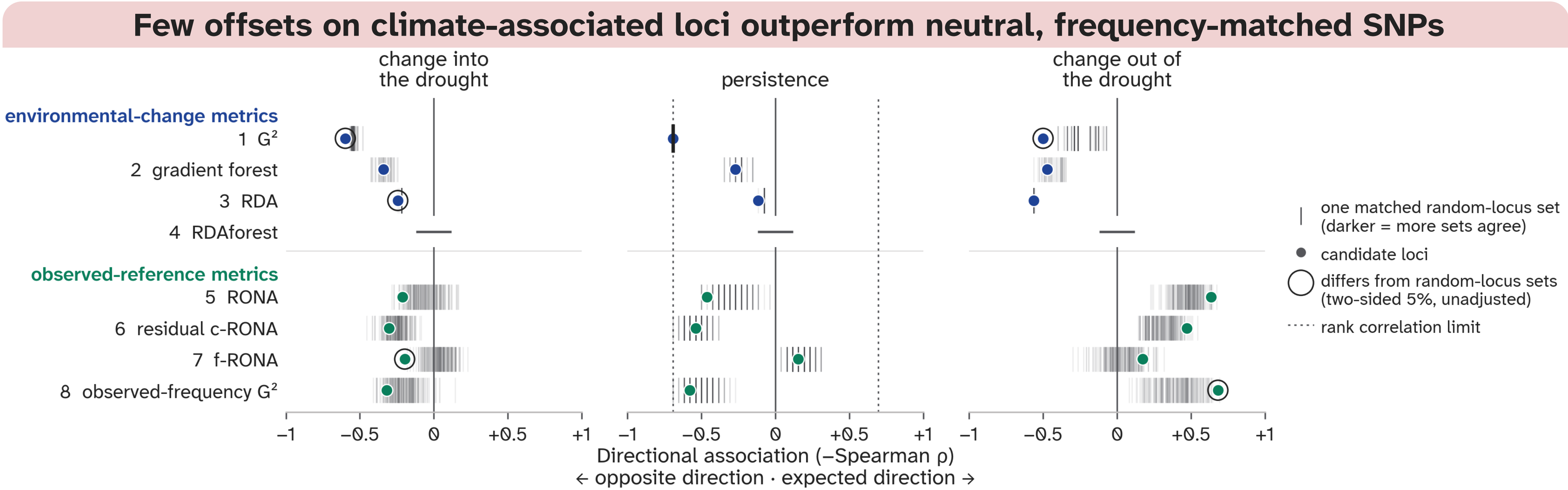


Geometric offset ranks the three extirpated populations as the safest



Methods

- Correlations:** We use **Spearman rank correlations** because we are interested in identifying the at-risk populations, not the specific fitted relationships.
- Effect sizes:** We report **-p**, such that positive values mean that populations ranked as riskier experienced worse outcomes.
- Covariates:** We use **partial Spearman rank correlations** to assess the impact of confounders like latitude, sample number, and sampling depth on effect size.
- Extirpation treatment:** We rank **extirpated populations together** at the worst rank on the 'change into drought' outcome.
- Random SNP benchmark:** We use **frequency-matched random draws of LD-pruned loci** for random-SNP versions of each predictor where applicable.



Takeaways

- Low offset doesn't mean low vulnerability** to climate events or change
- Benchmarking offset statistics** against climate distance and random SNPs can inform their interpretation
- Observed-frequency based offsets** should better account for existing maladaptation, but the overall contribution of genomics for these predictions remains unclear
- The **choice of outcome** to predict changes predictions significantly
- Developing even larger, longer-term datasets** will be essential for validating predictors of resilience in the field

When do genomes add information?

Moving forward (to see progress so far, scan the QR code):

- Compare genomic offset** alongside other genomic, demographic, and environmental predictors on a wide range of demographic outcomes
- Ask **when genomic data adds value** to predictions of conservation relevance
- Explain the performance** of various predictors and interpret the underlying biology
- Build a **generalizable pipeline** to evaluate climate resilience in other systems
- Build **SLiM simulations** (individual-based, forward-in-time) to evaluate the conditions under which our results do or do not hold